



PORTFOLIO CASE STUDY · US AIRLINE OPERATIONS

Flight Delay Contagion

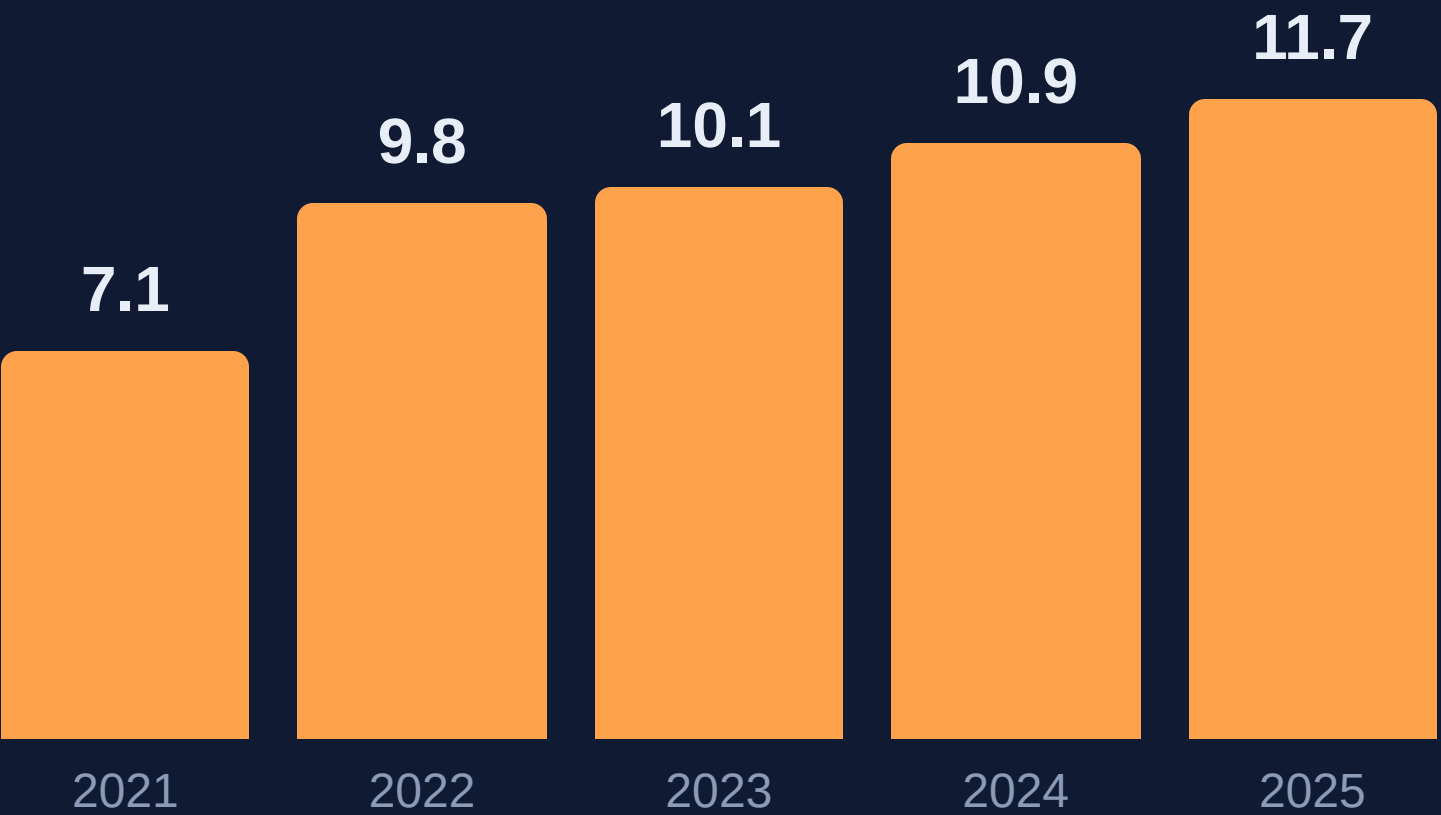
How one late plane ruins the whole day, and what it would take to stop it

Prepared for the VP of Operations, American Airlines

33.65M US domestic flights, 2021–2025 · PostgreSQL · Python · Excel · Power BI · github.com/lalithreddy335/flight-delay-contagion

Delays cost US airlines ~\$50B in five years, and it's getting worse

Estimated airline delay cost, \$B per year
(+64% since 2021)



FLIGHTS

33.65M

US domestic flights, Jan 2021 to Dec 2025

DELAY COST

\$49.6B

Five years, in 2025 dollars

ON-TIME, 2025

77.7%

Worst of the five years (82.8% in 2021)

LATE AIRCRAFT

38.8%

Of all delay minutes (BTS cause codes)

Source: BTS On-Time Performance, 2021–2025 · Delay cost = arrival delay minutes × \$98.41 per aircraft minute (Airlines for America, 2025)

Delays spread plane to plane, and late-morning delays spread most

How a delay chain works

Every flight is linked to its aircraft's previous leg by tail number

- 1 · Leg one: a delay starts
- 2 · Leg two inherits it
- 3 · Leg three inherits it again
- 4 · The overnight break ends it

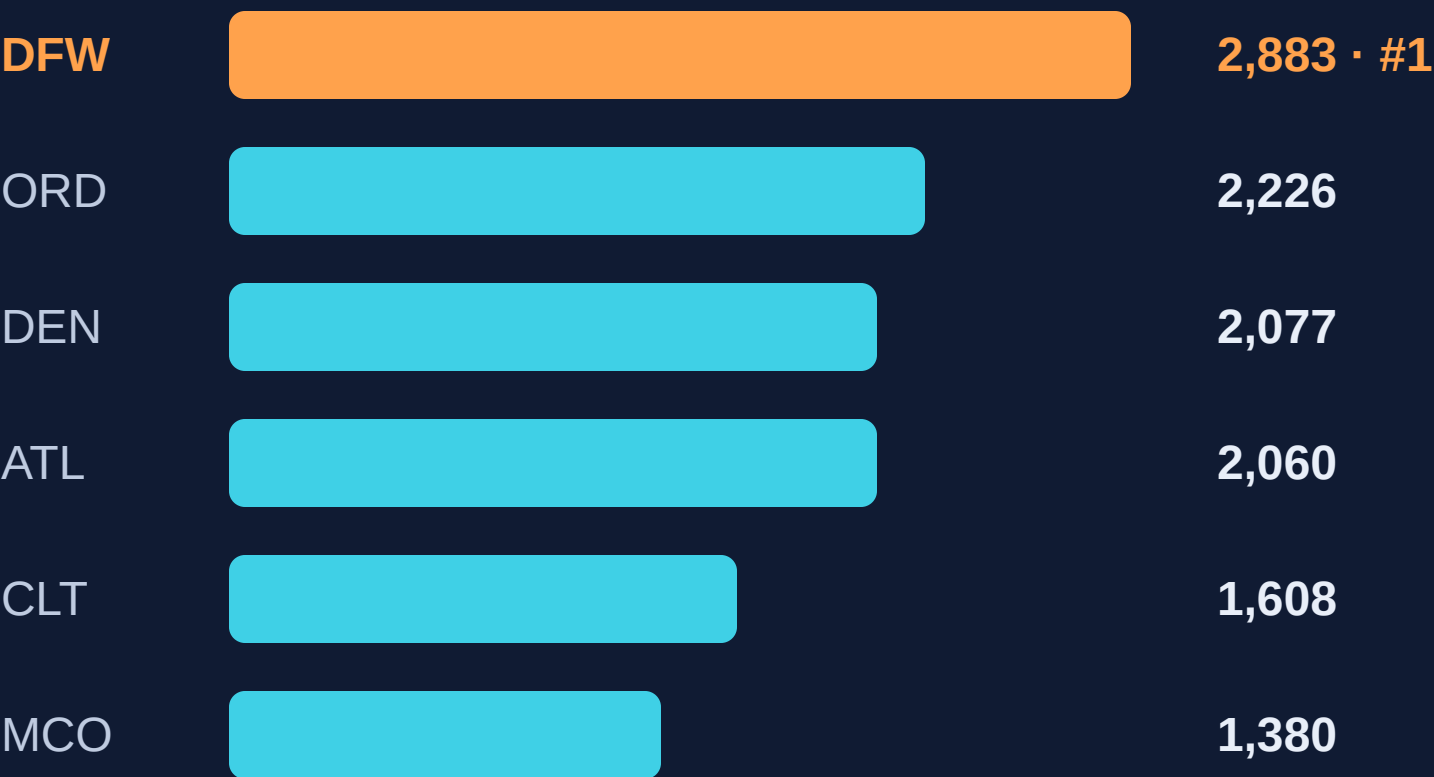
Chance a chain spreads to the next flight, by when the first delay starts



4.1M chains traced. 41.5% of chain delay minutes were inherited, close to the 38.8% late-aircraft share in BTS's own cause codes.

DFW starts the costliest chains, and it's American's biggest hub

Delay-chain cost by origin airport, \$M, 2021–2025



\$8.28B

American: highest chain cost of any US airline

64.0% vs 56.3%

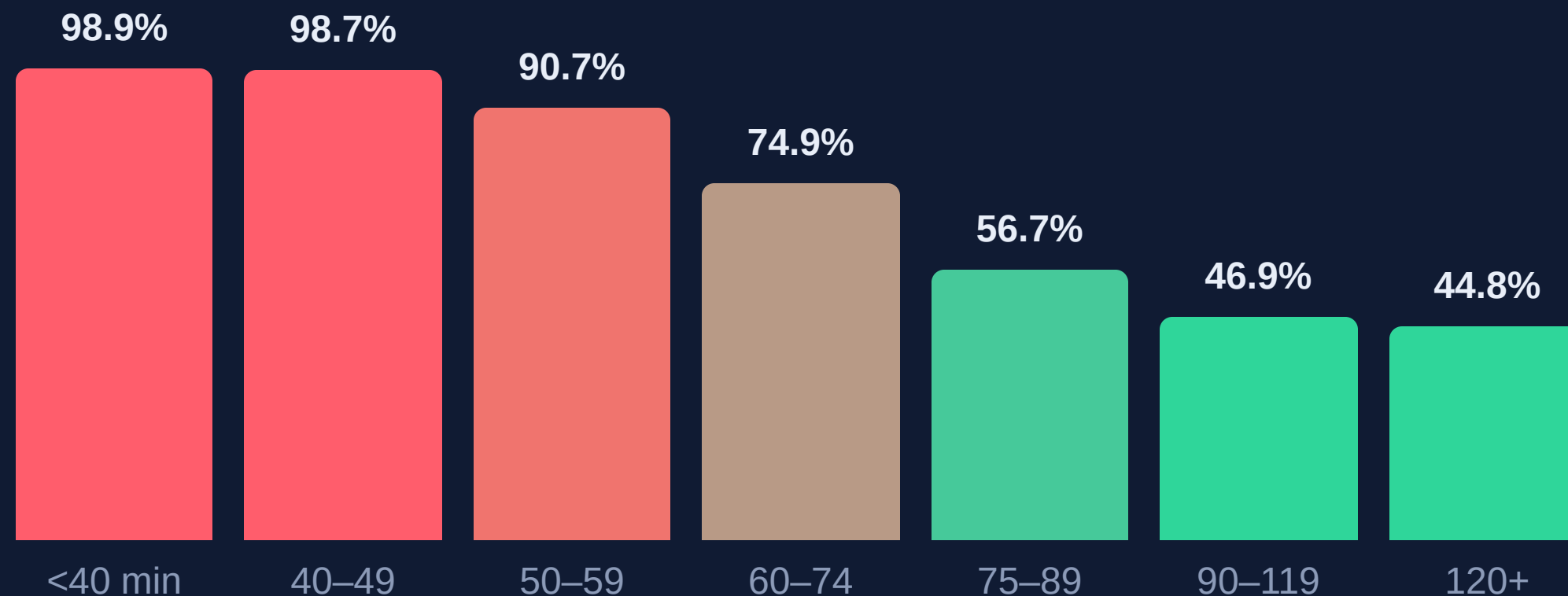
Delays passed on: AA at DFW vs Delta at ATL

68 vs 75 min

Median turnaround: AA at DFW vs Delta at ATL

More turnaround time stops delays from spreading

Plane arrived 15+ min late: % that left late too, by scheduled turnaround (AA at DFW)



0.57 min

of delay absorbed per extra buffer minute, controlling for incoming delay, time of day and year (95% CI 0.55–0.60)

1.9×

Turns under 60 min pass a late arrival's delay on 93.9% of the time, vs 49.1% at 75+ min (chi-square $p < 0.001$)

Blanket buffers lose \$20M a year. Targeted ones nearly break even.

Blanket 75-min rule

Every American turn at DFW raised to at least 75 min

Turns buffered per year	74,000
Buffer minutes added	1.31M
Delay minutes saved	121.6K
Break-even ground cost	\$9.17 / min

Net per year

-\$20.2M loss

Risk-targeted buffers

Only turns whose inbound flight was late 40%+ of the prior year

Turns buffered per year	6,109
Buffer minutes added	105K
Delay minutes saved	19.2K
Break-even ground cost	\$17.99 / min

Net per year

**-\$0.7M, loss cut
97%**

Recommendation: pilot targeted buffers, don't roll out blanket ones

01 Skip blanket buffers

A 75-minute rule on every DFW turn would lose about \$20M a year.

02 Pilot buffers on ~6,100 high-risk turns

Turns whose inbound flight was late 40%+ of the prior year: about 105K buffer minutes a year, roughly break-even on direct savings.

03 Fix chronically late inbound flights at the source

Buffering at DFW treats the symptom; the late inbound routes are the cause.

Measure before scaling: the pilot pays off if knock-on savings reach 1.37× direct savings. The chain analysis measured 1.31× for evening delays and 2.08× for afternoon delays.

Break-even multiplier = \$24.60 assumed ground cost ÷ \$17.99 targeted break-even

How it was built

01

Data

33.65M BTS flight records, 2021–2025, downloaded with Python and bulk-loaded into PostgreSQL

02

SQL

LAG() over each tail number links every flight to its aircraft's previous leg: 4.1M delay chains

03

Statistics

Chi-square, regression and a z-test in Python confirm the effects, all at $p < 0.001$

04

Model and dashboard

An Excel scenario model turns findings into dollars; Power BI tells the story in three pages

Validation: the chain logic found 41.5% of delay minutes inherited; BTS's own cause codes say 38.8%. The SQL and Excel savings figures match exactly.

The Power BI dashboard: one story in three pages



01 · The problem
~\$50B in delay costs, and rising



02 · Where chains start
DFW and American lead



03 · The fix
Targeted buffers and a pilot

Built on 11 summary tables in PostgreSQL, with a custom dark theme and designed page backgrounds